

1. What is the difference between Global and Local Scope of a Variable?

Ans:

Variables in Python can have either global scope or local scope depending on where they are declared.

Global Variable

Declared outside a function.

Can be accessed throughout the program.

Exists until the program ends.

Local Variable

Declared inside a function.

Can be accessed only within the function where it is defined.

Exists only while the function executes.

Example

```
x = 10    # Global variable

def my_function():
    y = 20    # Local variable
    print("Local Variable:", y)

my_function()
print("Global Variable:", x)
```

Output:

```
Local Variable: 20
Global Variable: 10
```

Example of Error

```
def my_function():
    y = 20
```

```
my_function()
print(y)
```

Output:

```
NameError: name 'y' is not defined
```

This happens because `y` is a local variable.

2. What is a Recursive Function?

A recursive function is a function that calls itself repeatedly until a specified condition (base case) is met.

Syntax

```
def function_name():
    if base_condition:
        return
    function_name()
```

Example: Factorial Using Recursion

```
def factorial(n):
    if n == 1:
        return 1
    return n * factorial(n - 1)
```

```
print(factorial(5))
```

Output:

```
120
```

How it Works

```
factorial(5)
= 5 × factorial(4)
= 5 × 4 × factorial(3)
= 5 × 4 × 3 × factorial(2)
= 5 × 4 × 3 × 2 × factorial(1)
= 5 × 4 × 3 × 2 × 1
= 120
```

Advantages of Recursion

1. Makes code shorter and easier to understand for some problems.
2. Useful for problems like factorial, Fibonacci series, tree traversal, and searching algorithms.

Important Note

Every recursive function must have a base condition; otherwise, it will call itself forever and cause a `RecursionError`.